TASK 4

Setup and Use a Firewall on Windows

Cybersecurity Internship Practical Report

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Operating System	Windows 11
Firewall Tool	Windows Defender Firewall with Advanced Security
Date	10 November 2026

1. Objective

The objective of this practical was to configure and test basic firewall rules that allow or block network traffic, using Windows Defender Firewall on Windows 11.

A custom inbound rule was created to block incoming TCP traffic on port 23 (Telnet). The configuration was verified, basic connectivity checks were run in PowerShell, and the test rule was removed to restore the previous firewall configuration.

2. Tools and Technologies Used

Tool / Technology	Purpose in this practical
Windows Defender Firewall	Built-in Windows 11 firewall that filters network traffic.
Windows Defender Firewall with Advanced Security	Graphical interface used to view, create, verify and delete firewall rules.
Windows PowerShell 5.1	Terminal used for connectivity and listener checks.
Test-NetConnection	PowerShell cmdlet used to test a TCP connection to a port.
Get-NetTCPConnection	PowerShell cmdlet used to check TCP connection and listening-port information.
ipconfig	Command used to inspect network adapter configuration.
System.Net.Sockets.TcpListener	.NET class used to create a temporary TCP listener.

3. Practical Environment

Item	Details
Operating System	Windows 11
Firewall	Windows Defender Firewall
Configuration Interface	Windows Defender Firewall with Advanced Security
Terminal	Windows PowerShell 5.1
Test Protocol	TCP
Test Port	23 (Telnet)
Rule Direction	Inbound
Rule Action	Block the connection

4. Detailed Step-by-Step Procedure

The steps below are listed in the order in which they were performed.

Step 1: Open Windows Defender Firewall

- Opened the Windows Start menu.
- Searched for Windows Defender Firewall.
- Opened the firewall configuration window.

Step 2: Access Advanced Firewall Settings

- Clicked Advanced Settings.
- Opened Windows Defender Firewall with Advanced Security.
- Selected Inbound Rules and examined the existing rules without modifying them.

Step 3: Create a New Inbound Rule

- Clicked New Rule in the Actions panel.
- Selected Port as the rule type and clicked Next.

Step 4: Configure Protocol and Port

Setting	Value
Protocol	TCP
Port selection	Specific local ports
Port number	23

Port 23 is commonly associated with Telnet.

Step 5: Configure the Firewall Action

Selected **Block the connection** to configure the rule to block incoming TCP traffic matching the specified port.

Step 6: Select Firewall Profiles

Kept all three profiles selected: Domain, Private and Public.

Step 7: Name and Save the Rule

Field	Value
Rule Name	Black Telnet Port 23
Description	Blocks incoming TCP traffic on port 23 to prevent Telnet connections.

Clicked Finish to create the rule. The name above is reproduced as entered during the practical.

Step 4. Detailed Procedure (continued)

Step 8: Verify the Firewall Configuration

Located the new rule in the Inbound Rules list, opened its Properties and checked the Protocols and Ports tab.

Property	Verified value
Protocol Type	TCP
Local Port	23
Remote Port	All Ports
Rule status	Enabled
Profiles	Domain, Private and Public

This step verified the rule's configuration only. It did not establish that actual network traffic was blocked.

Step 9: Initial Connectivity Test

Opened Windows PowerShell and ran the following command:

```
Test-NetConnection -ComputerName localhost -Port 23
```

Field	Result
ComputerName	localhost
RemotePort	23
PingSucceeded	True
TcpTestSucceeded	False

PingSucceeded: True indicates that the local host responded to the ping check. TcpTestSucceeded: False indicates that a TCP connection to port 23 could not be established. This result alone does not prove that the firewall blocked the connection. No application was listening on port 23 at the time, and a connection to a port with no listening service also fails. The result cannot distinguish between those causes.

Step 10: Check the Network Configuration

Ran **ipconfig** and recorded the following network information:

Adapter	IPv4 Address	Subnet Mask	Default Gateway
Ethernet 2	192.168.56.1	255.255.255.0	Not recorded
Wi-Fi	192.168.1.33	255.255.255.0	192.168.1.1

The Wi-Fi IPv4 address (192.168.1.33) was identified as the address that could be used for a network test.

4. Detailed Step-by-Step Procedure (continued)

Step 11: Check Whether Port 23 Was Listening

Ran the following command:

```
Get-NetTCPConnection -LocalPort 23 -State Listen -ErrorAction SilentlyContinue
```

No output was returned, which means no listening TCP service was detected on port 23 at that time. This supports the interpretation that the failed connection in Step 9 could be explained by the absence of a service.

Step 12: Prepare a Temporary TCP Listener

Entered the following commands in PowerShell to prepare a temporary TCP listener on port 23, bound to all local IPv4 addresses:

```
$listener = [System.Net.Sockets.TcpListener]::new([System.Net.IPAddress]::Any, 23)\n$listener.Start()\nWrite
```

Step 13: Stop the Temporary TCP Listener

Stopped the listener using the following commands:

```
$listener.Stop()\nWrite-Host "Temporary TCP listener stopped."
```

The listener was stopped after the test preparation. No connection attempt was performed while the listener was running.

Testing limitation

A remote connection test was not performed because another computer connected to the same Wi-Fi network was unavailable. Therefore, the practical did not conclusively demonstrate the firewall rule blocking a real incoming connection. The report distinguishes configuration verification from a successful traffic-blocking test.

Step 14: Remove the Test Firewall Rule

- Returned to Windows Defender Firewall with Advanced Security.
- Opened Inbound Rules and located the test rule named Black Telnet Port 23.
- Right-clicked the rule, selected Delete and confirmed the deletion.

The temporary rule was removed to restore the previous firewall configuration, as instructed in the internship task.

5. Commands Executed

Step	Command	Purpose
9	Test-NetConnection -ComputerName localhost -Port 23	Test a TCP connection to port 23 on the local computer.
10	ipconfig	Display IP configuration of network adapters.
11	Get-NetTCPConnection -LocalPort 23 -State Listen -ErrorAction SilentlyContinue	Check whether a TCP service is listening on port 23.
12	TcpListener constructor; $listener.Start(); Write-Host	Prepare a temporary TCP listener on port 23.
13	$listener.Stop(); Write-Host	Stop the temporary TCP listener.

5. Commands Executed (continued)

The commands below are reproduced in the order used during the practical.

Step 9: Connectivity test

```
Test-NetConnection -ComputerName localhost -Port 23
```

Step 10: Network configuration

```
ipconfig
```

Step 11: Listening-port check

```
Get-NetTCPConnection -LocalPort 23 -State Listen -ErrorAction SilentlyContinue
```

Step 12: Prepare the temporary TCP listener

```
$listener = [System.Net.Sockets.TcpListener]::new([System.Net.IPAddress]::Any, 23)\n$listener.Start()\nWrite
```

Step 13: Stop the temporary TCP listener

```
$listener.Stop()\nWrite-Host "Temporary TCP listener stopped."
```

The firewall rule itself was created, verified and deleted through the graphical interface, so no PowerShell commands were used for those steps.

6. Observations and Results

The firewall rule was created and its configuration was verified in the graphical interface. The connectivity checks were limited and did not demonstrate the rule blocking a real incoming connection.

No.	Observation	Type
1	Existing inbound rules were inspected without modification.	Configuration review
2	A custom inbound firewall rule was created.	Configuration
3	TCP port 23 was configured as the specific local port.	Configuration
4	Block the connection was selected.	Configuration
5	Domain, Private and Public profiles were selected.	Configuration
6	Properties showed TCP, local port 23, remote port All Ports, and enabled status.	Configuration verification
7	Initial test returned TcpTestSucceeded: False.	Connectivity check
8	No listening service was detected on port 23.	Connectivity check
9	A temporary TCP listener was prepared and subsequently stopped.	Test preparation and cleanup
10	The test firewall rule was deleted.	Cleanup

6. Observations and Results (continued)

6.1 Configuration Verification versus Network Traffic Testing

Aspect	What was done	What it shows
Configuration verification	Checked rule Properties (protocol, local port, remote port, status and profiles).	The rule was defined and enabled as configured.
Local connectivity check	Ran Test-NetConnection against localhost on port 23.	The TCP connection failed; the cause was not isolated because no service was listening.
Remote connection testing	Not performed.	No evidence was collected from a second computer.

6.2 Analysis of the Connectivity Results

- TcpTestSucceeded: False means a TCP connection to port 23 could not be established.
- Get-NetTCPConnection returned no output, so no service was listening on port 23 at that time.
- A connection attempt to a port with no listening service can fail even when no firewall block rule is present.
- The test used localhost, which refers to the same computer; it did not send traffic from another device over the network.
- A temporary TCP listener was prepared and later stopped, but no connection test was performed while it was running.

6.3 Limitations

A remote connection test from another computer on the same Wi-Fi network could not be performed because another computer was unavailable. A more conclusive test would require a listening service and a connection attempt from a second device, comparing the result with the firewall rule enabled and disabled. This comparison was not performed, so the blocking behavior was not demonstrated conclusively.

Screenshots were captured and shared during the practical session. They are not embedded in this revised report copy; they can be added separately as supporting evidence if required by the internship submission.

7. Technical Explanation

7.1 What a Firewall Is

A firewall is a security control that monitors network traffic and decides whether to allow or block it based on a set of rules. It helps protect a device or network by controlling which connections are permitted.

7.2 How a Firewall Filters Traffic

When a packet arrives, the firewall compares it with its rules. A rule can check details such as direction, protocol, local and remote ports, program and network profile. If a rule matches, its configured action is applied. Windows Defender Firewall uses default behavior and rule precedence to determine whether traffic is allowed or blocked.

7.3 Inbound and Outbound Traffic

Type	Meaning	Example in this practical
Inbound traffic	Connections initiated from another system and directed to this computer.	A device on the network attempting to connect to port 23 on this computer.
Outbound traffic	Connections initiated by this computer and directed to other systems.	This computer opening a web page.

The rule created in this practical was an inbound rule, so it was intended to apply to incoming connections.

7.4 Firewall Rules

A firewall rule is an instruction that specifies which traffic to match and what action to take. The test rule had the following configuration:

Property	Value
Direction	Inbound
Rule type	Port
Protocol	TCP
Local port	23
Action	Block the connection
Profiles	Domain, Private, Public

7.5 TCP Protocol

Transmission Control Protocol (TCP) is a connection-oriented transport protocol. It establishes a connection before data transfer and provides reliable, ordered delivery. TCP uses a three-way handshake to establish a connection. Test-NetConnection with the -Port option attempts a TCP connection and reports whether it succeeds.

7. Technical Explanation (continued)

7.6 Port Numbers

A port number (0 to 65535) identifies a service or application endpoint on a device. An IP address identifies the device, while the port number helps identify the destination service. A service generally needs to listen on a port to accept incoming TCP connections. If no service is listening, a TCP connection attempt will normally fail.

7.7 Port 23 and Telnet

Port 23 is the standard TCP port associated with Telnet, a protocol used for remote command-line access. Telnet does not encrypt its session by default, so information such as usernames and passwords can be exposed if traffic is captured. SSH is commonly used instead because it provides encryption. Blocking unnecessary Telnet access can reduce exposure to insecure remote access.

7.8 Domain, Private and Public Profiles

Profile	Used when	Typical network context
Domain	Connected to an organization's domain network.	Centrally managed organizational network.
Private	Connected to a network marked as trusted, such as a home or work network.	Trusted network.
Public	Connected to a public network, such as a cafe or airport.	Untrusted network; typically more restrictive.

The test rule was configured for all three profiles, so it was set to apply across Domain, Private and Public network profiles.

7.9 Windows Defender Firewall with Advanced Security

This is the advanced management console for Windows Defender Firewall. It allows users to view and manage inbound and outbound rules, create rules based on ports, programs or predefined services, select profiles, and inspect rule properties. It was used in this practical to create, verify and delete the test rule.

7. Technical Explanation (continued)

7.10 Purpose of Blocking Unnecessary Incoming Connections

Every network service that accepts connections can create an entry point that may be exposed to unwanted access. Blocking incoming connections to services that are unnecessary or insecure, such as Telnet, can reduce a system's attack surface and lower the risk of unauthorized access or exploitation. This supports the principle of least privilege: allow only the network traffic that is required.

8. Conclusion

This practical provided experience in using Windows Defender Firewall with Advanced Security on Windows 11. A custom inbound rule was configured to block incoming TCP traffic on port 23, and its settings were checked through the rule's Properties.

The main learning outcomes were:

- **Firewall configuration through the Windows GUI:** navigating from Windows Defender Firewall to the Advanced Security console and reviewing existing inbound rules without changing them.
- **Creating inbound rules:** using the New Rule wizard to choose the rule type, protocol, port, action, profiles, name and description.
- **TCP port configuration:** setting TCP and a specific local port (23) for a firewall rule.
- **Blocking incoming traffic:** selecting the Block the connection action for the test rule.
- **Checking connectivity with PowerShell:** using Test-NetConnection and Get-NetTCPConnection to check a port and whether a service was listening.
- **Inspecting network configuration:** using ipconfig to identify the Wi-Fi and Ethernet 2 IPv4 addresses and subnet masks.
- **Restoring the configuration:** deleting the temporary rule after the practical to return the firewall to its previous configuration.

The practical also demonstrated why a failed connection test must be interpreted carefully. TcpTestSucceeded: False does not by itself prove that a firewall rule blocked traffic; a port with no listening service can also cause a connection attempt to fail. A remote test was not possible because another computer on the same Wi-Fi network was unavailable. Therefore, the firewall configuration was completed and verified, but its blocking behavior was not conclusively demonstrated through a remote traffic test.

End of Report